

Robert Jankowski

Quantum & Computer Engineering Department, TU Delft
research@robertjankowski.net | robertjankowski.github.io

Professional Experience

Postdoctoral Fellow

Nov 2025–present

TU Delft

Topic: Shortest paths in large incomplete networks

Advisor: Maksim Kitsak

Education

Ph.D. in Network Geometry

2021–2025

University of Barcelona

Thesis: *Hyperbolic Cartography of Complex Networks*

Supervisors: M. Ángeles Serrano and Marián Boguñá

Visiting Researcher, Luddy Center for Artificial Intelligence, Indiana University, Aug–Dec 2024

Hosts: Santo Fortunato and Filippo Radicchi

M.Sc. in Data Exploration and Interdisciplinary Modeling

2020–2021

Warsaw University of Technology

Thesis: *Interactions in signed complex networks*

Supervisor: Piotr Górski

B.Sc. in Applied Physics

2016–2020

Warsaw University of Technology

Thesis: *Predicting election results from opinion polls using machine-learning methods*

Supervisor: Julian Sienkiewicz

Publications

Peer-reviewed

L. B. Rosselló, **R. Jankowski**, H. Bersini, M. Boguná, M. Á. Serrano. *Trade-offs between structural richness and perceptual robustness in music network representations*. Phys. Rev. Research, (accepted, 2026)

R. Aliakbarisani*, **R. Jankowski***, M. Á. Serrano, M. Boguná. *HypBench: Hyperbolic Benchmark for Graph Neural Network Performance*. IEEE Transactions on Neural Networks and Learning Systems (2026)

A. F. Marcús, **R. Jankowski**, M. V. Miñana, C. Casacuberta, M. Á. Serrano. *Chordless cycle filtrations for dimensionality detection in complex networks via topological data analysis*. Nature Communications, **17**, 6105 (2026)

R. Jankowski, F. Radicchi, M. Á. Serrano, M. Boguná, S. Fortunato. *Task complexity shapes internal representations and robustness in neural networks*. Machine Learning: Science and Technology, **7** 025045 (2026)

G. Bernárdez, L. Telyatnikov, M. Papillon, . . . , A. Weers, **R. Jankowski**, L. Cino, . . . , N. Miolane. *Topological Deep Learning Challenge 2025: Expanding the Data Landscape*. Proceedings of the 1st Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2025), Proceedings of Machine Learning Research **321**, 4–14 (2026)

M. Kim, L. Cirigliano, C. Castellano, H. Sun, **R. Jankowski**, A. Poggialini, F. Radicchi. *Shortest-path percolation on scale-free networks*. Physical Review E, **113**, 014314 (2026)

R. Jankowski, R. Aliakbarisani, M. Á. Serrano, M. Boguná. *Mapping bipartite networks into multidimensional hyperbolic spaces*. Communications Physics **9**, 35 (2026)

R. Jankowski, P. Hozhabrierdi, M. Boguná, M. Á. Serrano. *Feature-aware ultra-low dimensional reduction of real networks*. npj Complexity **1**, 13 (2024)

R. Jankowski, A. Allard, M. Boguná, M. Á. Serrano. *The D-Mercator method for the multidimensional hyperbolic embedding of real networks*. Nature Communications **14**, 7585 (2023)

R. Jankowski, A. Chmiel. *Role of Time Scales in the Coupled Epidemic-Opinion Dynamics on Multiplex Networks*. Entropy **24**, 105 (2022)

R. Jankowski, J. Sienkiewicz. *Determining Crucial Factors for the Popularity of Scientific Articles*. Acta Physica Polonica A **138**, 41–47 (2020)

*Equal contribution.

Preprints / Under Review

M. V. Miñana, **R. Jankowski**, A. F. Marcús, R. Ballester, M. Á. Serrano, C. Casacuberta. *Motif-based filtrations for persistent homology: A framework for graph isomorphism and property prediction*. arXiv:2604.15265.

Grants and Awards

Grants

External collaborator, YOUNG PW II grant “Application of the low-dimensional representation of real networks for localization of information source” (with Robert Paluch), Warsaw University of Technology, 2024–2026 (150,000 PLN).

Bridge grant with Lluc Bono Rosselló, Young Researchers of the Complex Systems Society (yrCSS), 2024.

Individual Funding

AccelNet-MultiNet program, 2024.

Predoctoral grant FI-SDUR, 2022–2025.

SECS scholarship (yrCSS), 2022.

Awards

Best Contributed Talk, Network Science Informs AI Satellite at NetSci, 2026.

Best Student Presentation, SFINKS Conference, 2021.

Best engineering thesis in Poland, Section Physics in Economy and Social Sciences of Polish Physical Society, 2021.

Rector’s scholarship for academic performance: 2020 (3rd/317), 2019 (top 5), 2018.

Talks and Posters

Invited Talks

Network geometry and multidimensional hyperbolic maps of real networks.

Faculty of Physics, Warsaw University of Technology, Poland, 2024

Contributed Talks

Task complexity shapes internal representations and robustness in neural networks.

Network Science Informs AI (NSIA) Satellite at NetSci 2026, Boston, USA, 2026

IAIFI PhD Summer School, Boston, USA, 2025

Network representations reveal structured uncertainty in music.

XIII Polish Symposium on Physics in Economics and Social Sciences, Warsaw, Poland, 2025

Mapping bipartite networks into multidimensional hyperbolic spaces.

NetSci 2025, Maastricht, The Netherlands, 2025

Feature-aware ultra-low dimensional reduction of real networks.

NetSci 2024, Quebec City, Canada, 2024

The D-Mercator method for the multidimensional hyperbolic embedding of real networks.

APS March Meeting, Minneapolis, USA, 2024

2nd Meeting of the Spanish Chapter of the CSS, Barcelona, Spain, 2024

Statphys28, Tokyo, Japan, 2023

NetSci 2023, Vienna, Austria, 2023

CCS 2022, Palma de Mallorca, Spain, 2022

Role of time scales in coupled epidemic-opinion dynamics on multiplex networks.

NetSci-X 2022, virtual/Porto, Portugal, 2022

The influence of relations in forming interactions among communities on social websites.

SFINKS conference, 2021. Best student presentation

Crucial factors determining the popularity of scientific articles.

10th Polish Symposium on Physics in Economy and Social Sciences, 2019

Posters

Benchmarking Hyperbolic Graph Embeddings for Link Prediction and Topology Reconstruction.

NetSci 2026, Boston, USA, 2026

HypBench: Hyperbolic Benchmark for Graph Neural Network Performance.

Non-Euclidean Foundation Models and Geometric Learning Workshop @ NeurIPS 2025, San Diego, USA, 2025

Task complexity shapes internal representations and robustness in neural networks.

IAIFI Summer Workshop, Boston, USA, 2025

Feature-aware ultra-low dimensional reduction of real networks.

3rd Meeting of the Spanish Society of Complex Systems, Madrid, Spain, 2025

D-Mercator: multidimensional hyperbolic embedding of real networks.

XXIV Congreso de Física Estadística, Pamplona, Spain, 2023

Generating interactions for friendship networks.

NetSci 2023, Vienna, Austria, 2023

From relation to interactions: a case study in Reddit website.

11th Polish Symposium on Physics in Economy and Social Sciences, 2021

Teaching and Outreach

M.Sc. thesis supervisor, TU Delft (2026). Topic: *Shortest Paths and Hyperbolic Geometry.*

Teaching Fellow, DIS Master Class: Network Renormalization, CSH Vienna, May 2025.

Delivered the workshop “Les xarxes complexes com a eina per resoldre problemes” at the University of Barcelona’s Festa de la Ciència (2023 and 2024).

Professional Service

Chair for Web and Design, Network Science Society (2026–present).

Invited Editor, Special Collection on “Network Science for Understanding AI,” *Physical Review E*.

Journal Reviewer: *Physical Review E*, *Communications Physics*, *Chaos*, *Scientific Reports*, *PLOS ONE*, *EPJ Data Science*, *Physica A*, and *Proceedings of the Royal Society A*.

Program Committee Member: NetSci (2024–2026), CompleNet (2026), ACM KDD (2024), and MISDOOM (2022).

Conference and Workshop Organization

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2026, Boston, MA, USA.

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2025, Maastricht, Netherlands.

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2024, Quebec, Canada.

Summer Schools and Workshops

IAIFI PhD Summer School & Summer Workshop

Institute for AI and Fundamental Interactions, Harvard University, USA

Aug 2025

Complementarity in Complex Networks Workshop

Delft, The Netherlands

May 2025

LOGML Summer School

London, United Kingdom

Jul 2024

Visualizing Complexity Science Workshop Complexity Science Hub, Vienna, Austria.	Aug 2023
XI GEFENOL Summer School on Statistical Physics of Complex Systems Barcelona, Spain	Jul 2023
Complex Networks: Theory, Methods, and Applications Summer School Lake Como School of Advanced Studies, Italy	May 2022

Industry and Research Experience

Research Intern Samsung R&D Institute Poland. Developed and evaluated automatic speech recognition systems as part of the Bixby team.	Jul–Aug 2021
Student Researcher Warsaw University of Technology. Group of Anna Chmiel. Project “Modeling epidemic spread using comorbidities and social attitudes”.	May–Dec 2021
Student Researcher Warsaw University of Technology. Group of Janusz Hołyst. Project ALPHORN on signed relations and structural balance (with ETH Zürich).	2020–2021
Research Intern Samsung R&D Institute Poland. Developed and tested deep-learning models for named entity recognition.	Jul–Sep 2020
Scala Intern & Junior Scala Developer TouK, Warsaw Technologies: <i>Scala, Akka, Python, Docker, RabbitMQ</i> .	Jul–Nov 2019
Software Engineering Intern Ilabo, Warsaw. Developed web applications in C# and .NET Core and performed data analysis in Python.	Jul–Aug 2018

Art Exhibitions

Capillary Network. Ars Electronica Festival 2025, Linz, Austria.

Capillary Network. Altering Nature: Exploring Life in Computational Art, HKUST (Guangzhou), China, April 2025.

Languages

Polish – Native.

English – Full working proficiency.

Spanish – Professional working proficiency.